

Saurabh Raj Varma

AI Engineer | Machine Learning Engineer | Data Scientist



About Me

Aspiring AI Engineer and Data Scientist with expertise in Machine Learning, Deep Learning, NLP, LLMs, and Full-Stack AI development. Passionate about building intelligent, scalable AI applications.

Personal

Name: Saurabh Raj Varma
Location: Chhatarpur, Madhya Pradesh
Education: B.Tech, Civil Engineering
Institute: IIT (ISM) Dhanbad

Areas of Specialization

Machine Learning • Deep Learning • NLP
LLMs • Computer Vision • Data Analytics

Programming

Python • SQL • JavaScript
React.js • FastAPI
TensorFlow • Scikit-learn
Pandas • NumPy
XGBoost • LightGBM
Git • Linux • Power BI • LaTeX

Achievements

• Kaggle Expert
• Bronze Medal – AI Mathematical Olympiad
• Silver Medal – SPR 2026 Mammography

Interests

Badminton • Chess
Yoga • Competitive Programming

EXPERIENCE/ INTERNSHIPS

May 2025-Jul 2025

AI & Strategy Development Intern

MITTI LABS · Mumbai

Leveraged Python and data analytics to support AI-driven decision-making in agricultural technology. Assisted in developing data pipelines, analyzing farmer and operational datasets, automating reporting workflows, and generating business insights to improve product adoption and operational efficiency.



Aug 2024-Sept 2024

Data Scientist Intern

YHILLS PVT. LTD. · Remote

Developed a time-series forecasting pipeline using Python and Machine Learning on 50,000+ stock market records. Improved forecasting accuracy by approximately 20% through data preprocessing, feature engineering, feature selection, and model optimization.



EDUCATION

2022–2026

Bachelor of Technology (B.Tech) in Civil Engineering

INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES),
DHANBAD · Dhanbad, Jharkhand
CGPA: 7.15/10.



2014–2021

Senior Secondary (Class XII) & Secondary (Class X)

JAWAHAR NAVODAYA VIDYALAYA, NOWGONG · Chhatarpur, Madhya Pradesh
Class XII: 88.2% (2021) | Class X: 85.6% (2019).



PROJECTS

AI Voice Chatbot

Gemini, FastAPI, Next.js

Engineered a full-stack AI voice assistant integrating speech-to-text, Gemini LLM, and text-to-speech for real-time conversations. Designed scalable FastAPI REST APIs and an interactive Next.js frontend, improving response efficiency and user experience.

Fraud Detection System

Python, Scikit-learn, Pandas, NumPy

Developed a machine learning fraud detection system on highly imbalanced transaction data. Leveraged Pandas and NumPy for preprocessing, SMOTE for class balancing, and ensemble learning to improve fraud detection accuracy and model robustness.

Mammography Report Classification

Python, PyTorch, Pandas, NumPy, Matplotlib, Kaggle

Developed a deep learning pipeline for automated mammography report classification using advanced preprocessing, visualization, and transfer learning techniques, optimizing model accuracy, robustness, and diagnostic performance on medical imaging datasets.

LANGUAGES

Hindi
English
Punjabi
Japanese

Native (Mother Tongue)
Professional
Fluent
Beginner (Learning)

ENGAGEMENTS

- **Kartavya:** Mentored 50+ students; improved academic performance by 30%.
- **Srijan:** Coordinated logistics for a 10,000+ attendee cultural festival.
- **Coding Club:** Participated in competitive programming and technical discussions.

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